

Experiment 16: Compare Different Classification Algorithms and Evaluate Their Performance

Aim

To compare different classification algorithms and evaluate their performance using **Accuracy**, **Precision**, **Recall**, **F1-Score**, and **Confusion Matrix**.

Python Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.naive_bayes import GaussianNB

# Load Dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Models
models = {
    "Logistic Regression": LogisticRegression(max_iter=200),
    "KNN": KNeighborsClassifier(n_neighbors=3),
    "Decision Tree": DecisionTreeClassifier(),
    "Random Forest": RandomForestClassifier(),
```

```

    "Naive Bayes": GaussianNB()
}

print("Algorithm Comparison\n")

for name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)

    print(name)
    print("Accuracy =", round(accuracy * 100, 2), "%")
    print("-" * 30)

```

Sample Output

```

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KNN

=====

```

Accuracy : 1.0

Precision: 1.0

Recall : 1.0

F1-Score : 1.0

Confusion Matrix:

```
[[19 0 0]
```

```
[ 0 13 0]
```

```
[ 0 0 13]]
```

```

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Logistic Regression

```

Accuracy : 1.0

Precision: 1.0

Recall : 1.0

F1-Score : 1.0

Confusion Matrix:

[[19 0 0]

[0 13 0]

[0 0 13]]

Decision Tree

Accuracy : 1.0

Precision: 1.0

Recall : 1.0

F1-Score : 1.0

Confusion Matrix:

[[19 0 0]

[0 13 0]

[0 0 13]]

SVM

Accuracy : 1.0

Precision: 1.0

Recall : 1.0

F1-Score : 1.0

Confusion Matrix:

[[19 0 0]

[0 13 0]

[0 0 13]]